(iii) Seasonality, or seasonal factor, captures the effect of seasons on demand.

$$\text{Seasonality, } S_i = \frac{\text{actual demand}}{\text{regressed, deasonalized demand}}$$

$$S_i = \frac{D_i}{\bar{D}_i} \longrightarrow 1(a)$$

or

$$D_i = S_i \bar{D}_i \longrightarrow 1(b)$$

(iv) Re-seasonalization (or Forecasting)

The regressed de-seasonalized demand, $\bar{D}_i$, $(i = 1, 2, \cdots t)$ must be re-seasonalized in order to obtain the forecasts

$$F_i \underset{1(b)}{=} S_i \bar{D}_i \longrightarrow (2)'$$

[D_i replaced by F_i]

In practice, we use an average seasonal factor, $\bar{S}_i$, in place of S_i

$$F_i \underset{(2)'}{=} \bar{S}_i \bar{D}_i \longrightarrow (2)$$

TIM 172 B, Lecture # 5 (1/20/26)

0. Complete stages in Forecasting (static)
(done; see above)

1. Static Forecasting Process: Theory & Practice

2. HW #2
   - Extend the submission date to
     Friday, (1/23/26); midnight - 1 minute

1. Static Forecasting Process : Theory & Practice

Given : Historical demand data, $(D_1, D_2, \ldots D_t)$
present

Determine (predict, forecast) future demand

$$(F_{t+1}, F_{t+2}, \ldots)$$

Process:

Step 1 : Lay out the given data in the first 4 columns of the spreadsheet

Remember to PLOT!

col. (1) Year	(2) Qtr.	(3) Period, i	(4) actual demand, D_i
1	1	1	D_1
	2	2	D_2
	3	3	D_3
	4	4	D_4
2	1	5	D_5
	2	6	⋮
	3	7	⋮
	4	8	⋮
3	1	9	⋮
⋮	⋮	⋮	

Step 2 : De-seasonalization : de-seasonalize the given demand data.

De-seasonalization is

- the process of compressing the data before it can be regressed
- to compress the data, we take a moving average over "p" data points, where p is the periodicity of the data, i.e., # of periods or seasons in each cycle of the data

$$p \triangleq \text{periodicity of the data}$$

(for weather, 1 cycle = 1 year, p = 4 seasons or periods)

Periodicity, p, can be odd (p = 3, 5, 7, ...) or even (p = 4, 6, 8, ...)

Case 1 : Periodicity is odd

$p = 3, 5, 7 \ldots$

Example $\quad p = 3 \Rightarrow$ moving average over 3 data pts.

Work from first principles

$$\underbrace{D_1 \quad D_2 \quad D_3}\quad D_4$$

De-seasonalized demand, $\bar{D}_2' = \dfrac{D_1 + D_2 + D_3}{3} \rightarrow (1)$

Similarly, $\bar{D}_3' = \dfrac{D_2 + D_3 + D_4}{3}$

For any general p, odd $(p = 3, 5, 7 \ldots)$

$$\bar{D}_t' = \left[\sum_{i = t - \lfloor p/2 \rfloor}^{i = t + \lfloor p/2 \rfloor} D_i \right] \Big/ p$$

divided by

Check $t = 2$, $p = 3 \Rightarrow \lfloor p/2 \rfloor = \lfloor 1.5 \rfloor = 1$

$$\bar{D}_2' = \left[\sum_{i=(2-1)}^{i=(2+1)} D_i \right] \Big/ 3 = \frac{D_1 + D_2 + D_3}{3},$$

which is eqn.(1): "It checks" $\Rightarrow$ "checking your work"

Case 2 : Periodicity is even
($p = 2, 4, 6, \dots$)

Example : $p = 4$ (HW #2, Prob. 4, worked example in the text)

$\Rightarrow$ 4 pt. moving average

$$D_1 \quad D_2 \quad D_3 \quad D_4 \quad D_5$$

$$\bar{D}'_{2.5} \quad \bar{D}'_{3.5}$$

$$\bar{D}'_3$$

$$\bar{D}'_{2.5} = \frac{D_1 + D_2 + D_3 + D_4}{4} \longrightarrow (1)$$

$$\bar{D}'_{3.5} = \frac{D_2 + D_3 + D_4 + D_5}{4} \longrightarrow (2)$$

$$\bar{D}'_3 = \frac{1}{2}\left[\bar{D}'_{2.5} + \bar{D}'_{3.5}\right] \longrightarrow (3)$$

$$\bar{D}'_3 \underset{(3,1,2)}{=} \left[\frac{D_1 + 2D_2 + 2D_3 + 2D_4 + D_5}{(4)(2)}\right] \rightarrow (4)$$

Eqn. (4) generalizes to

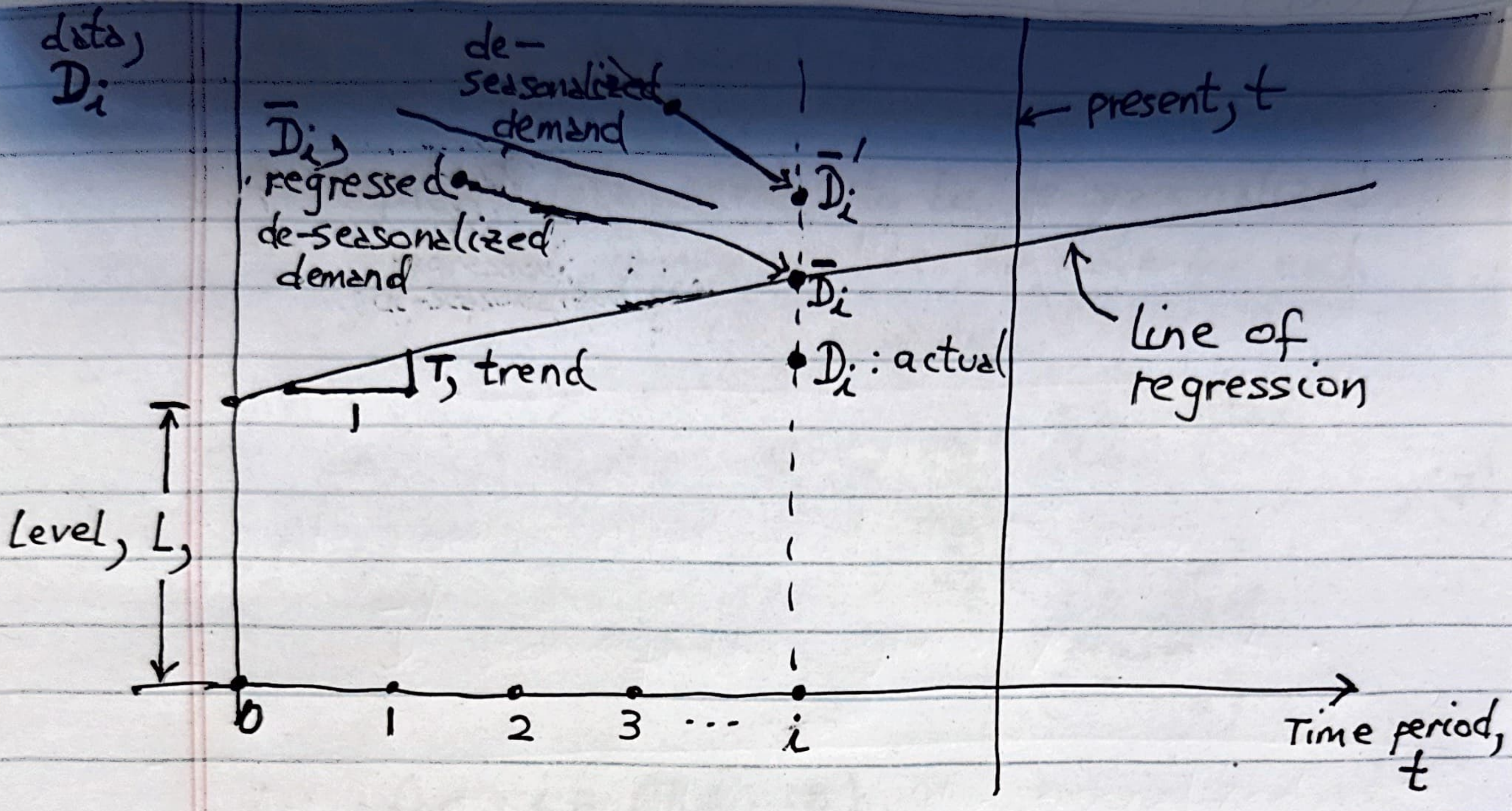
$$\bar{D}'_t = \frac{\left[D_{t-p/2} + D_{t+p/2} + 2\left(\sum_{i=t-(p/2-1)}^{i=t+(p/2-1)} (D_i) \right) \right]}{(2)(p)} \rightarrow (5)$$

At home, check that $t=3$, $p=4$, $\bar{D}'_3$ from eqn (5), gives you eqn(4).

$\Rightarrow$ Action step : Place the de-seasonalized demand data, $\bar{D}'_i$, calculated using the above eqns., in column 5 of the spreadsheet.

(Plot ......)

<u>Step 3</u> : Regress the de-seasonalized demand data, $\bar{D}'_i$, to obtain the regressed de-seasonalized demand, $\bar{D}_i$



$D_i \triangleq$ actual demand at time i, $(i = 1, 2, \ldots t)$
(TIME-SERIES)

$\bar{D}_i' \triangleq$ de-seasonalized demand, $(i = 1, 2, \ldots)$

$\bar{D}_i \triangleq$ regressed de-seasonalized demand, $(i = 1, 2,)$

Regression is performed on the de-seasonalized demand data pts, $\bar{D}_i'$, to obtain $\bar{D}_i$ given by

$$\bar{D}_i = L + iT \, , \quad (i = 1, 2, \ldots\ldots t) \rightarrow (6)$$

Action: Place the regressed de-seasonalized demand in column 6 of the spreadsheet

Step 4 : Seasonal factors

The data needs to be re-seasonalized. In order to re-seasonalize the data for each period in a cycle, compute

$$\text{Seasonal factor, } S_i = \frac{D_i}{\bar{D}_i} \quad \rightarrow (7)$$

D_i ← actual demand, (column 4)

$\bar{D}_i$ ← regressed deseasonalized demand (column 6)

for $i = (1, 2, \cdots p)$,
where $p \triangleq$ # of periods in each cycle of the data

⟹ Action : Place the seasonal factors, S_i, in column 7 of the spreadsheet

Step 5 : Calculate the average seasonal factors, $\bar{S}_i$, averaged over the number of cycles of available data.

for the worked example in the text (HW#2, Prob. 4), $n = 3$ cycles of data are available, & $p = 4$

$$\bar{S}_1 = \frac{S_1 + S_5 + S_9}{3}$$ [corresponding pts for each season]

$$\bar{S}_2 = \frac{S_2 + S_6 + S_{10}}{3}$$

$\bar{S}_3 = \dots$

$\bar{S}_4 =$

complete this at home

⇒ Action : Place the average seasonal factors in column (8) of the spreadsheet

<u>Step 6</u> : Re-seasonalization ≡ Forecasting